EXPLORATORY DATA ANALYSIS
RETAIL ORDERS DATASET

Task 2 Report | Python • Pandas • Matplotlib • Seaborn

Dataset: cleaned_retail_orders.csv | 1,000 rows × 17 columns

Purpose

Investigate statistical characteristics, distributions, relationships, category performance and monthly trends, then develop practical recommendations.

1. Executive Summary

This report explores 1,000 retail orders across 17 columns. It includes descriptive statistics, missing-value checks, distribution plots, correlation analysis, category comparisons, monthly trends, scatter plots and sensitivity analysis. Twenty flagged records (2% of orders) contribute 83.0% of gross order value. They are not confirmed errors and require verification.

Gross order value = Quantity × Unit Price before discounts. Estimated order value applies the recorded discount; missing discounts are omitted from the estimate total.

2. Objectives and Method

EDA was performed to summarize numeric data, inspect missingness, understand skewness, explore relationships, compare categories/months and produce business recommendations.

3. Dataset Overview and Missing Values

Column	Missing	Missing %
Coupon Code	880	88.0%
Customer Rating	261	26.1%
Discount %	97	9.7%
Discount Clean	97	9.7%
Estimated_Order_Value	97	9.7%
Payment Method	65	6.5%
Gender	43	4.3%
Customer Name	36	3.6%
City	29	2.9%
Order Date	29	2.9%

Customer Rating has 261 missing values (26.1%); Order Date has 29 (2.9%); Discount Clean has 97 (9.7%). Coupon Code is missing in 88% of rows but the cleaned coupon field labels these as “NO COUPON RECORDED”. Missing values should not automatically be treated as zero.

4. Descriptive Statistics

Mean is the average; median is the middle value; standard deviation measures spread; quartiles show the 25th, 50th and 75th percentiles. Median is less sensitive to extreme values than mean.

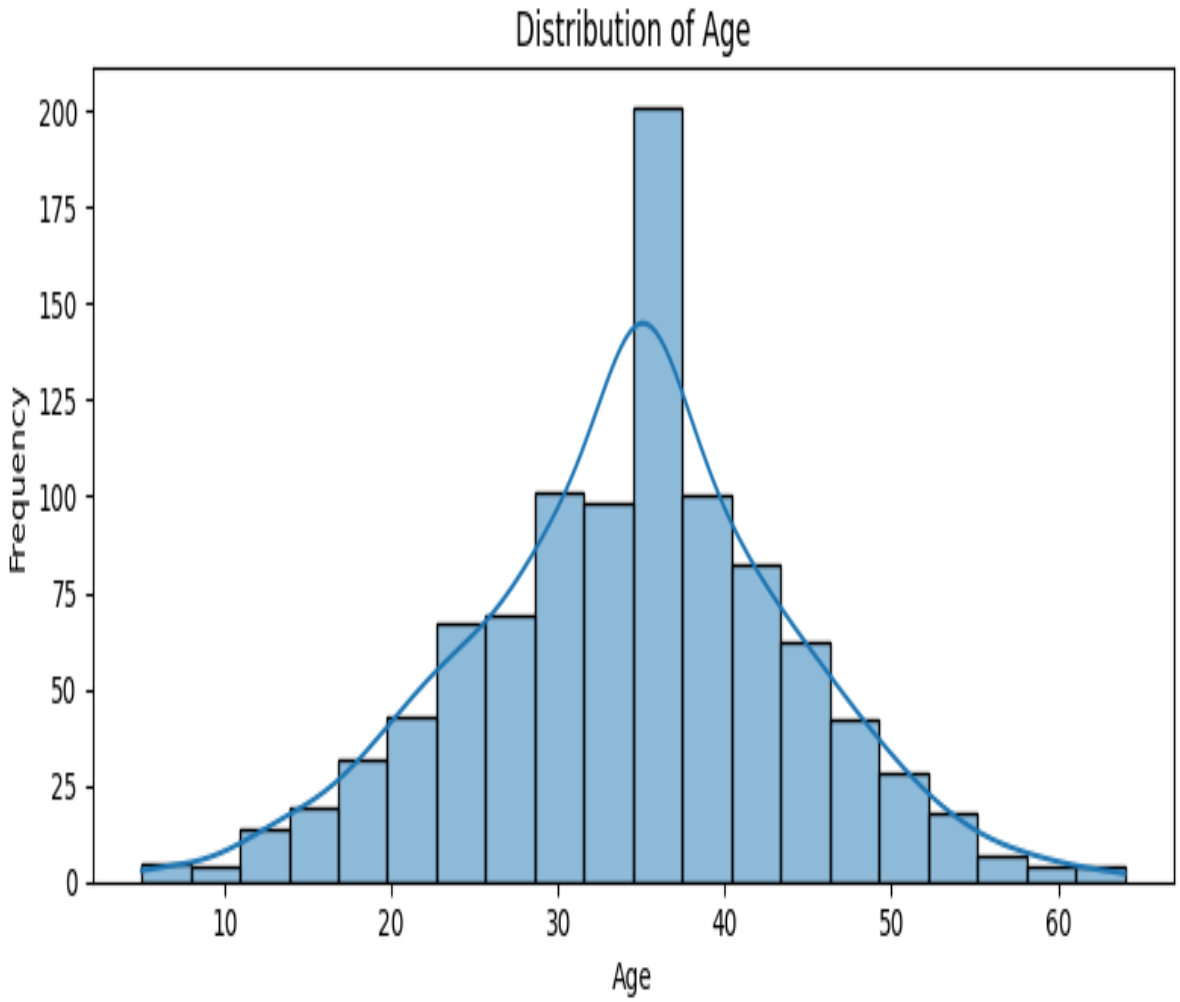
Variable	Count	Mean	Std	Min	Q1	Median	Q3	Max
Age	1,000.00	34.30	9.78	5.00	28.00	35.00	40.00	64.00
Quantity	1,000.00	5.20	20.86	1.00	2.00	4.00	6.00	500.00
Unit Price (INR)	1,000.00	9,548.24	82,689.98	134.27	633.49	1,098.64	1,837.26	1,200,000.00
Discount Clean	903.00	10.01	6.88	0.00	5.00	10.00	15.00	20.00
Customer Rating	739.00	3.75	1.13	1.00	3.00	4.00	5.00	5.00

Age median: 35; quantity median: 4; unit-price median: ■1,098.64. Maximum quantity is 500 and maximum unit price is ■1,200,000.00. These large maxima indicate strong skew or extreme observations.

5. Histograms and KDE Plots

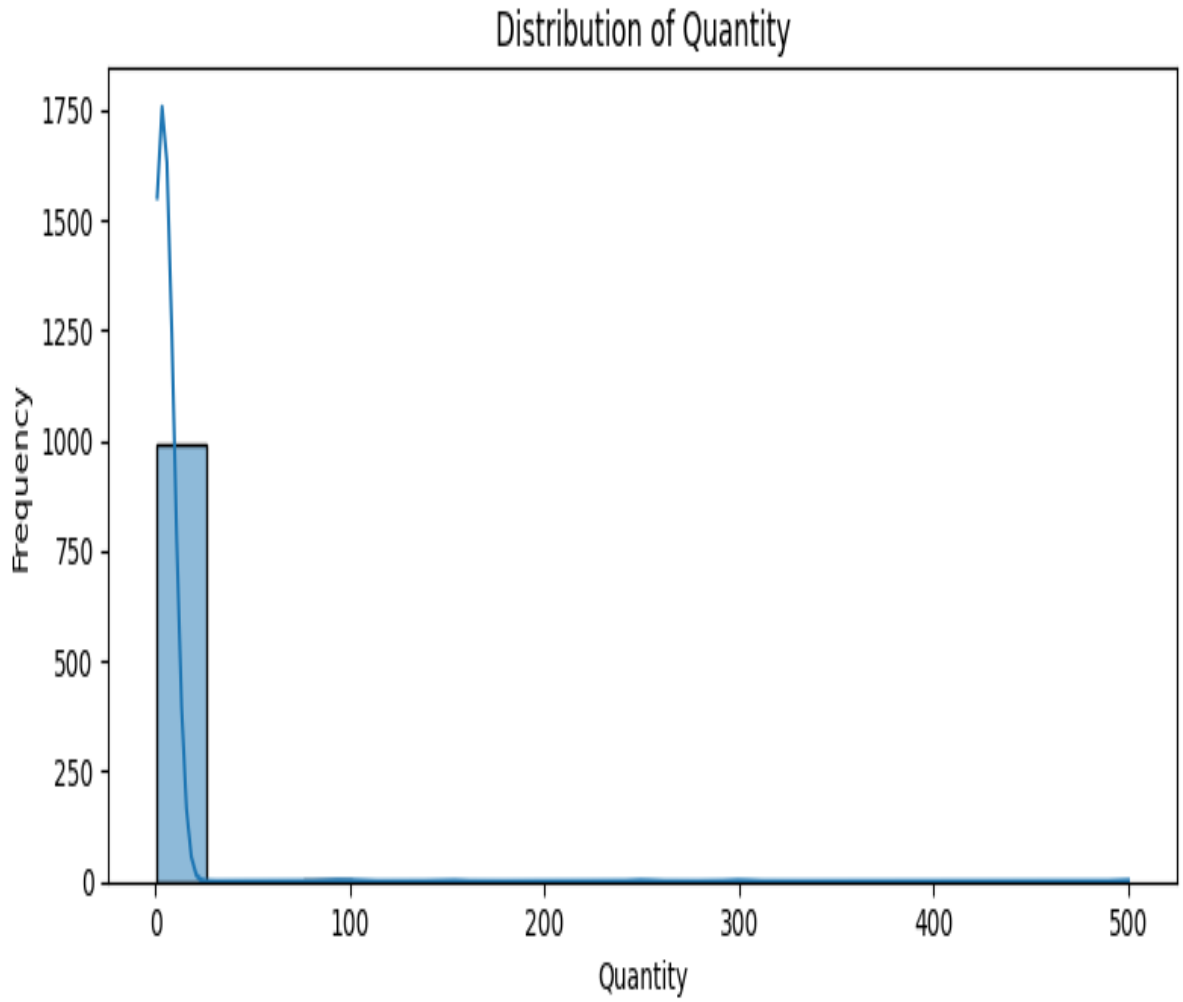
Histograms show frequency within ranges; KDE curves provide smoothed distribution shapes. KDE is less informative for discrete variables such as ratings and discount levels.

Age Distribution



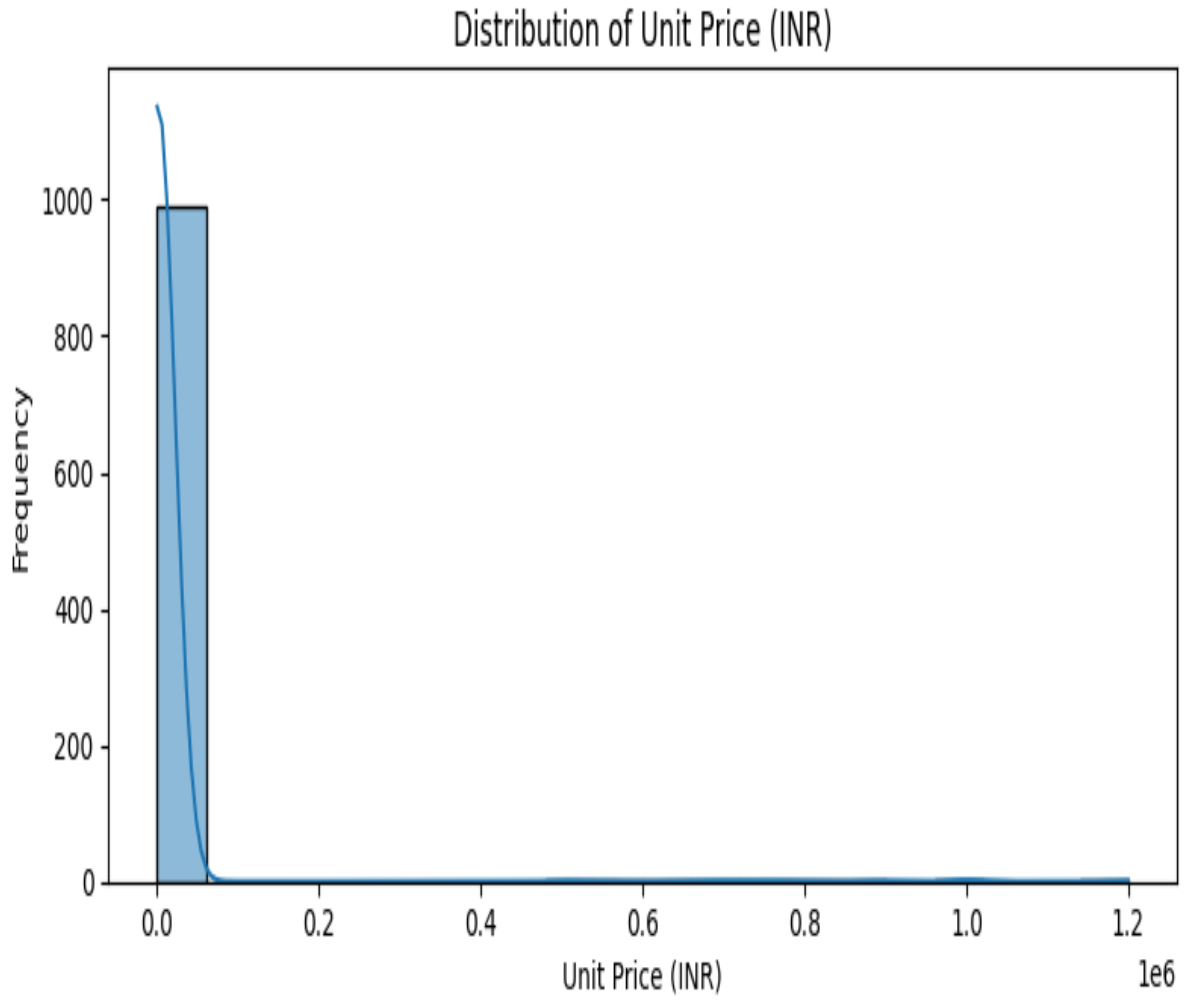
Why/analysis: Ages cluster around the 30s; median age is 35 and range is 5–64.

Quantity Distribution



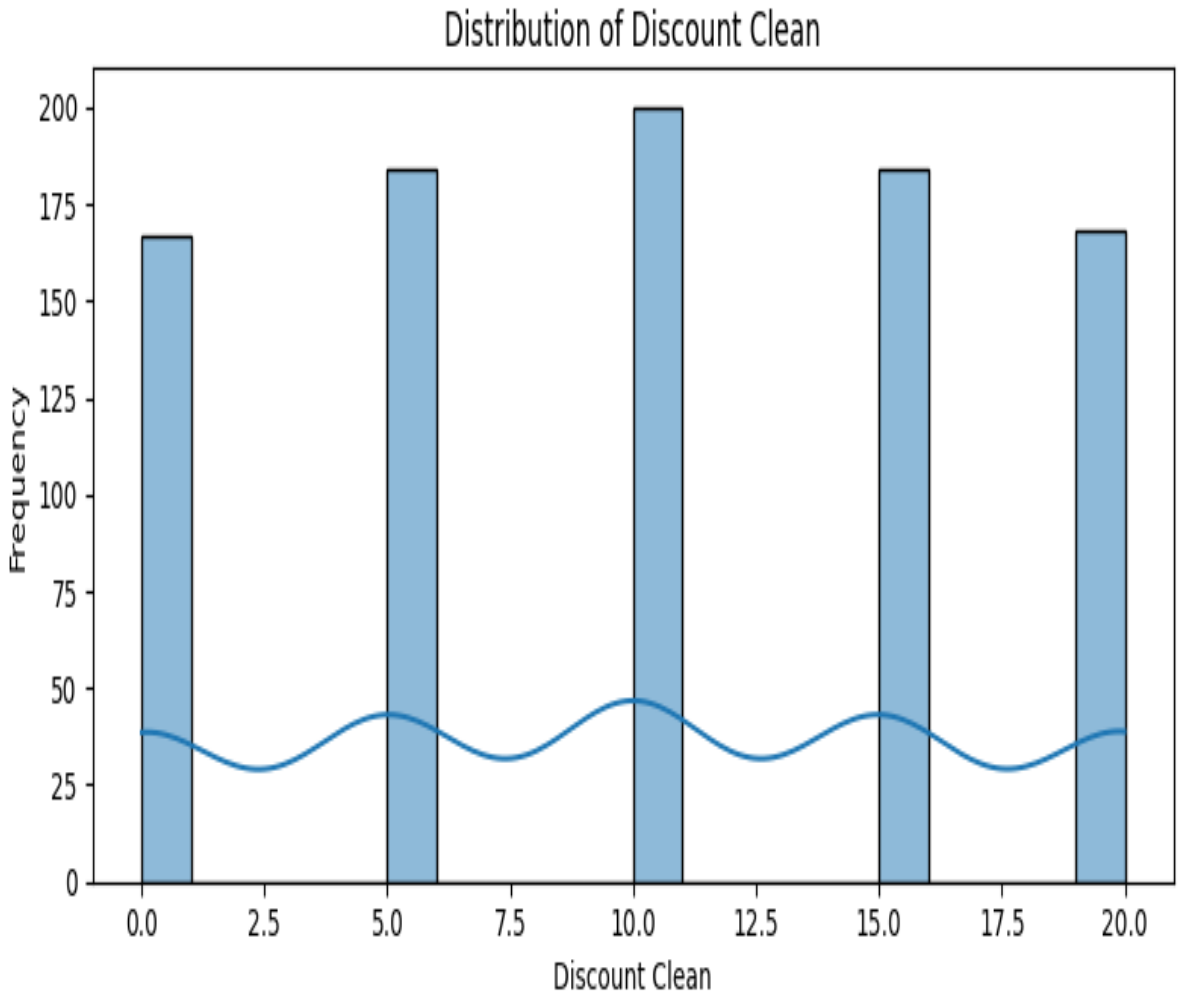
Why/analysis: Most orders have small quantities; a few very large orders create a long right tail and raise the mean.

Unit Price Distribution



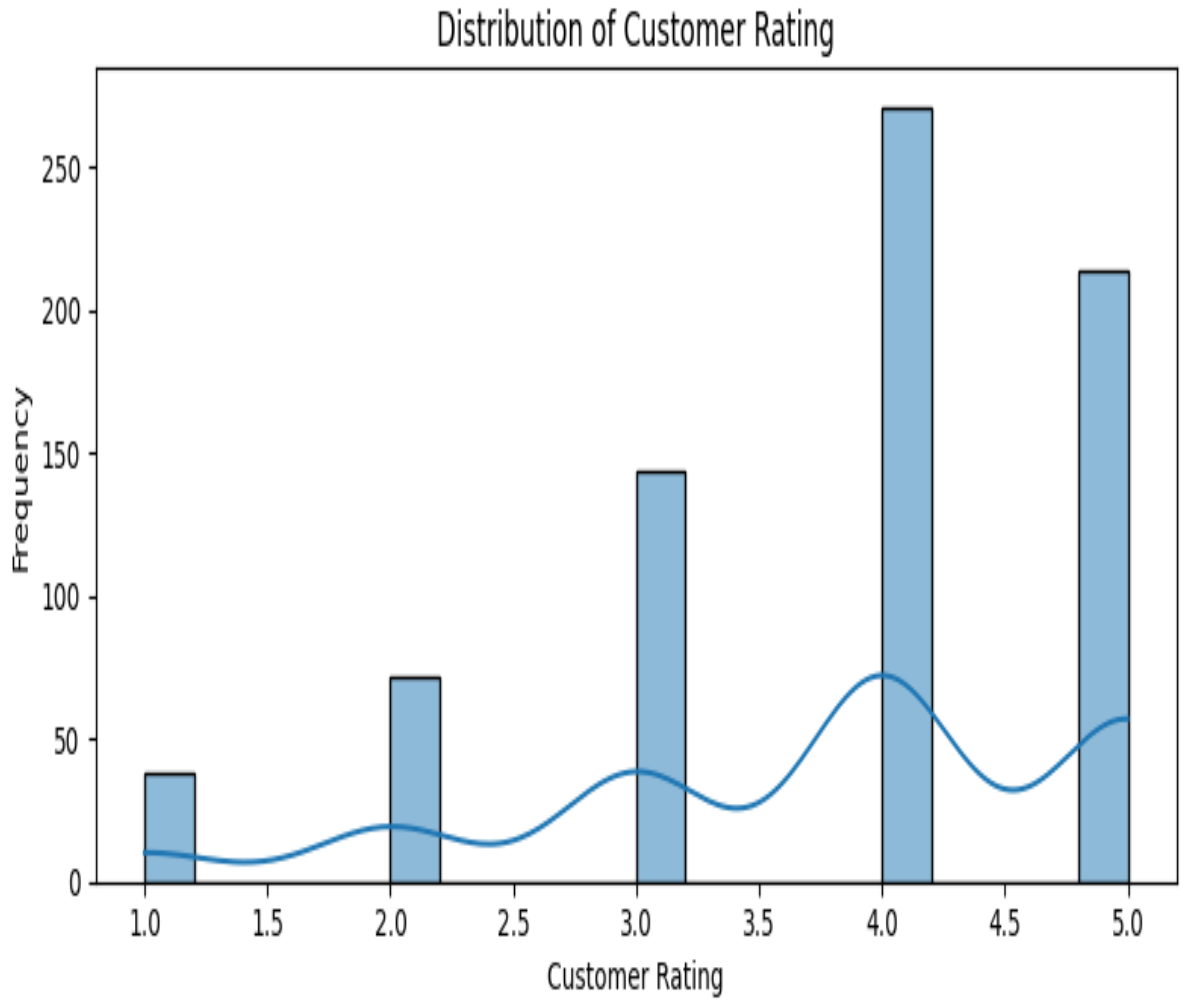
Why/analysis: Most prices are at the lower end, while extreme values stretch the axis to ■12,00,000; the median is more representative than the mean.

Discount Distribution



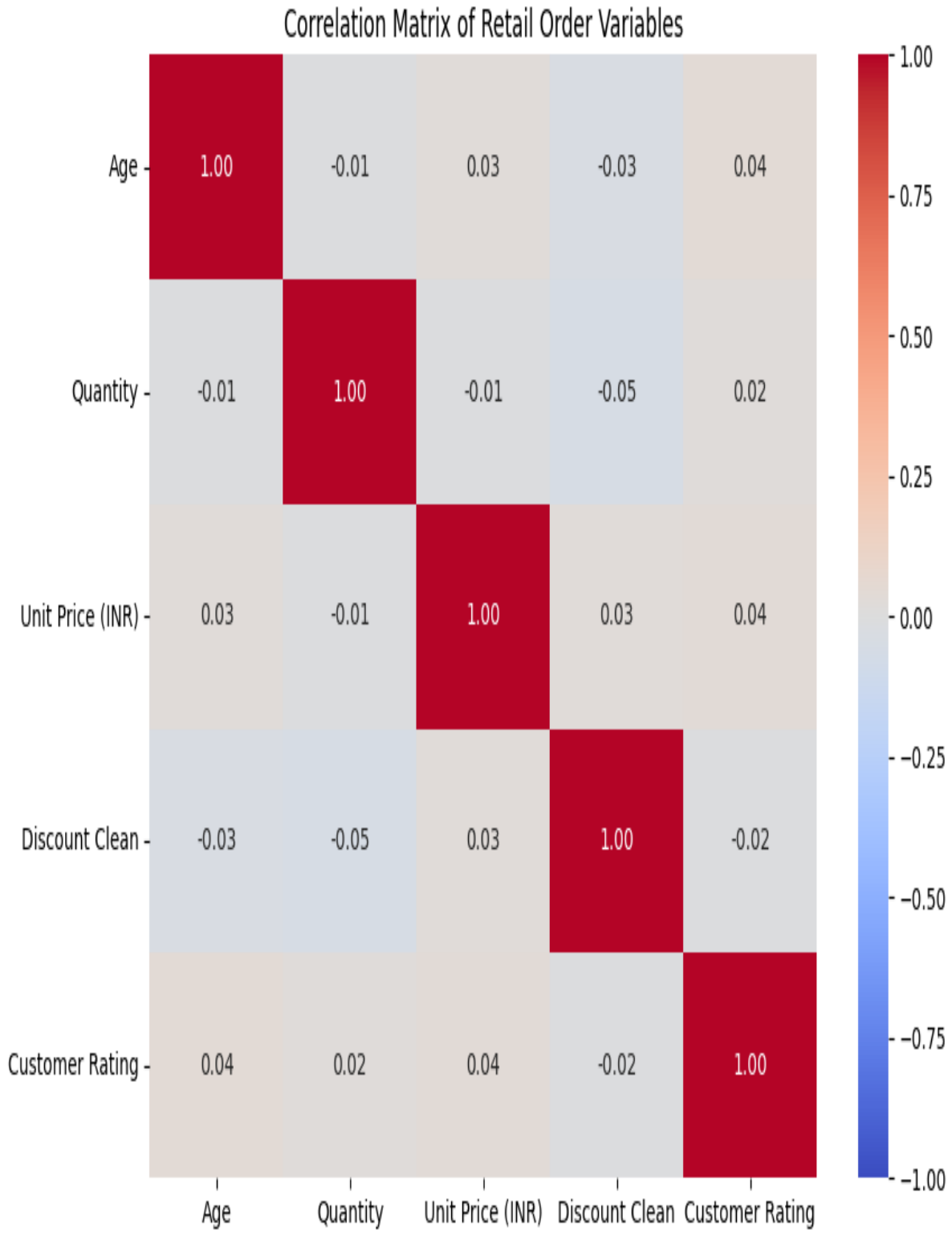
Why/analysis: Discounts occur at 0%, 5%, 10%, 15% and 20%; interpret KDE cautiously for these discrete values.

Customer Rating Distribution



Why/analysis: Ratings 4 and 5 are frequent among recorded responses, but 26.1% of ratings are missing.

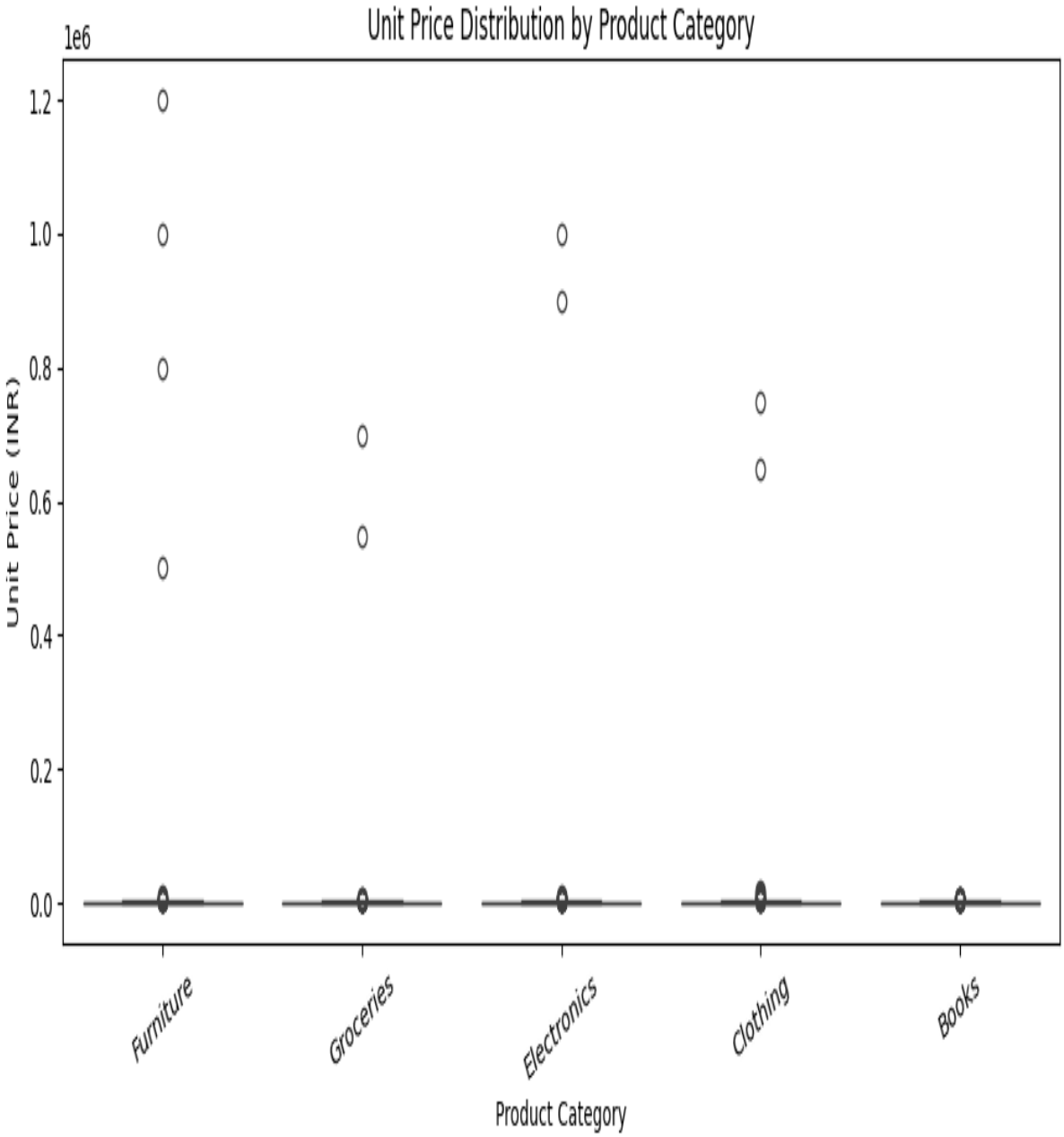
6. Correlation Matrix



Off-diagonal coefficients are close to zero (about –0.05 to +0.04 in the supplied output), indicating little linear association among selected numerical variables. Correlation does not prove causation; missing data and extreme values can affect coefficients.

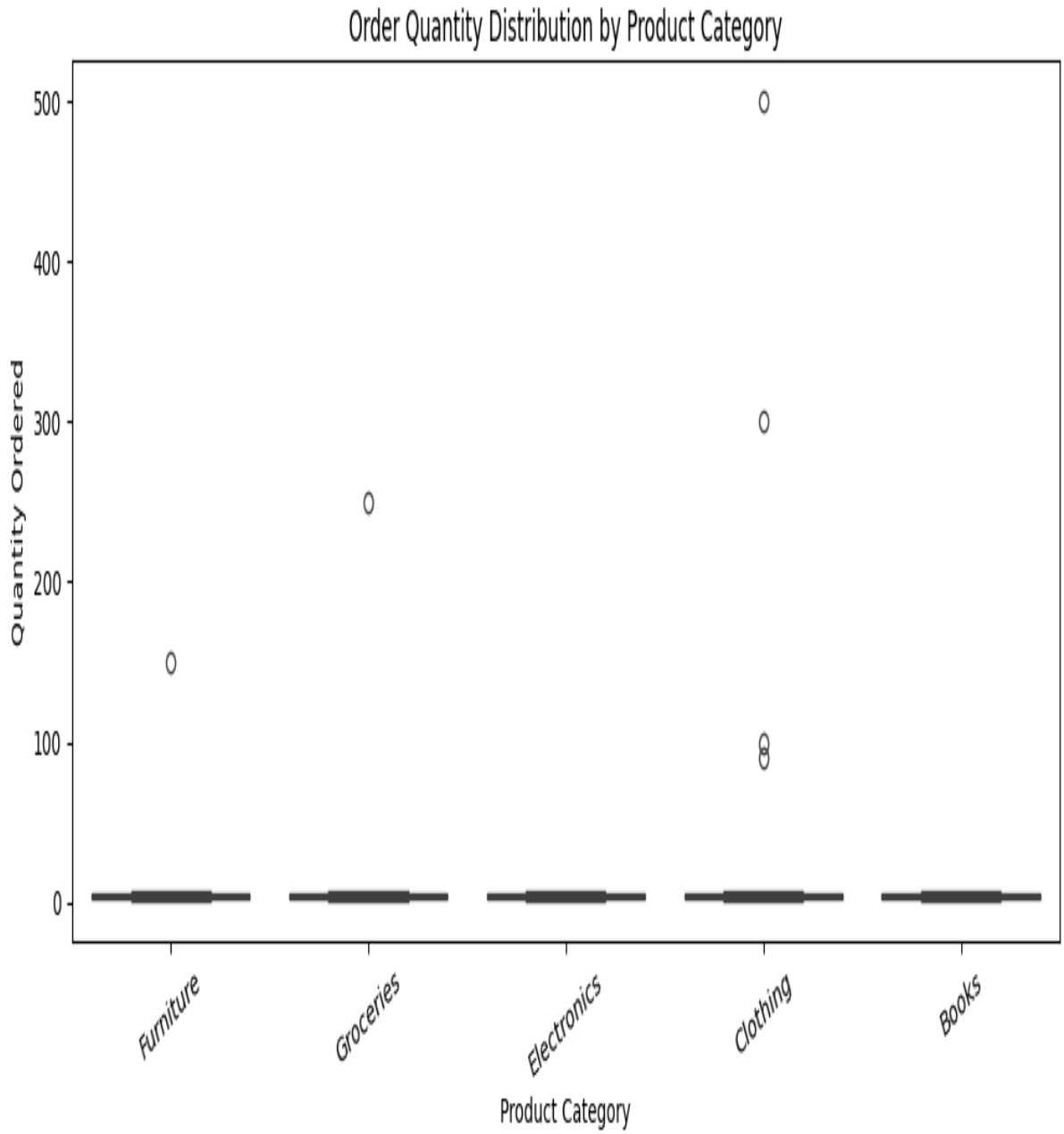
7. Box Plots and Category Comparisons

Unit Price by Product Category



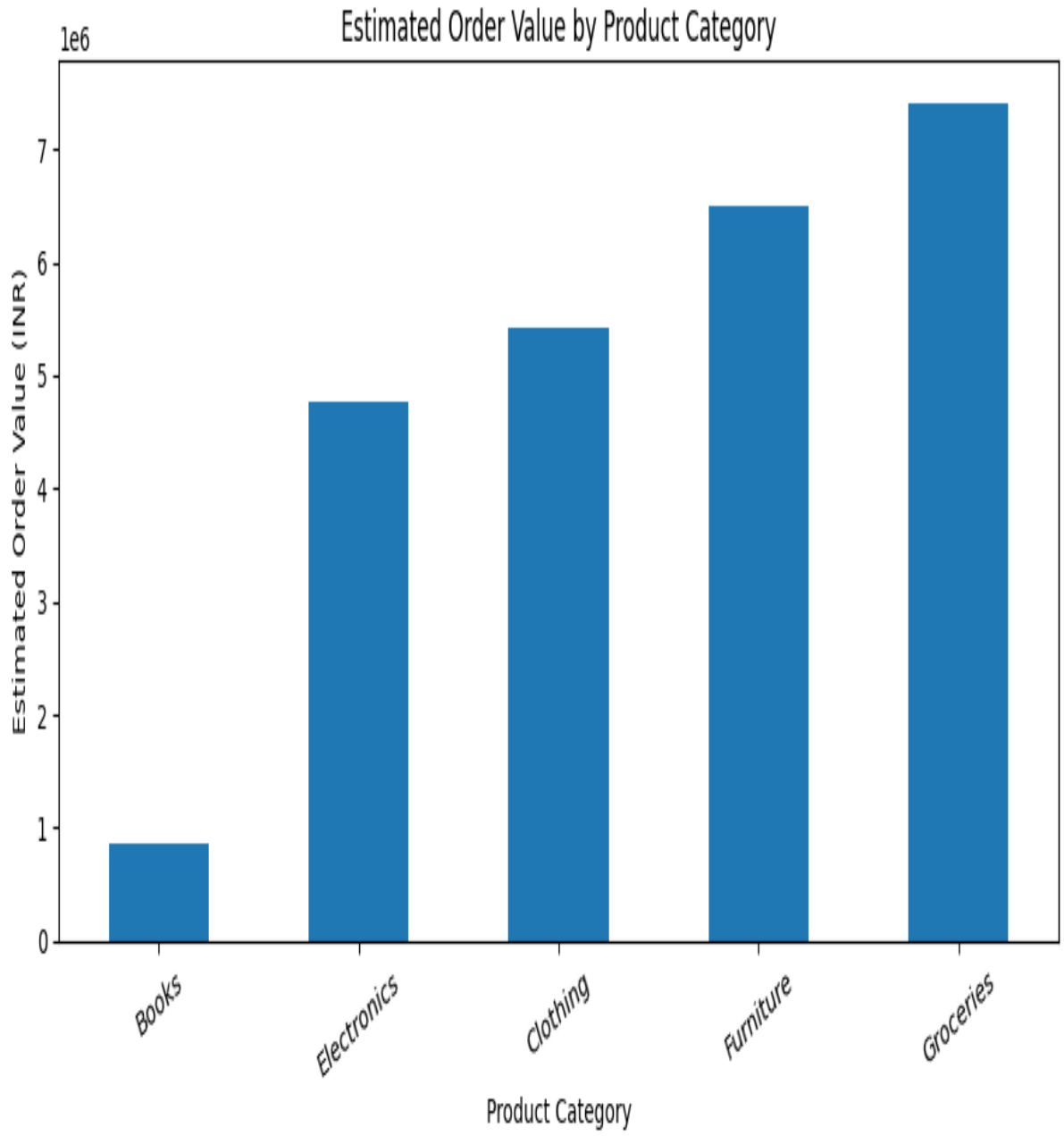
High-price outliers compress the central boxes. Validate extreme prices and consider log-scale or typical-range charts.

Quantity by Product Category



Large quantities appear as outliers, especially in Clothing. These may be bulk orders or errors and should not be deleted without validation.

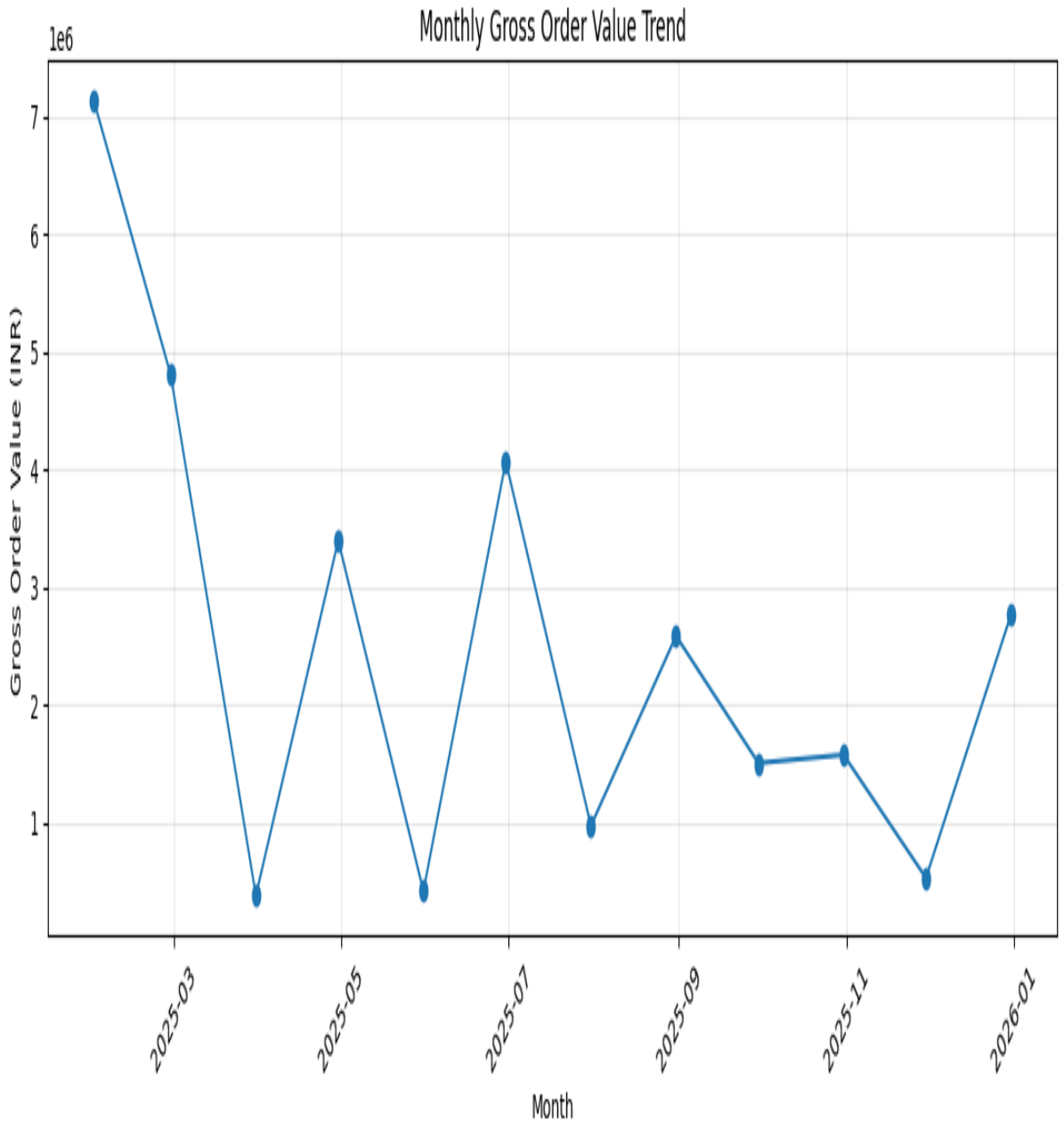
Estimated Order Value by Product Category



Category	Orders	Total Qty	Avg Qty	Median Price	Estimated Value
Groceries	194	1011	5.21	■983.07	■7,408,604.87
Furniture	210	920	4.38	■1,172.58	■6,495,742.71
Clothing	212	1805	8.51	■1,142.47	■5,429,203.93
Electronics	191	731	3.83	■1,054.91	■4,766,919.01
Books	193	733	3.80	■1,096.87	■851,396.89

The supplied summary showed Groceries with highest estimated value (about ■74.09 lakh), Books lowest (about ■8.51 lakh). Clothing had the most orders (212) and highest total quantity (1,805); Furniture had highest median unit price (■1,172.58). Totals are preliminary due to extreme values and missing discounts.

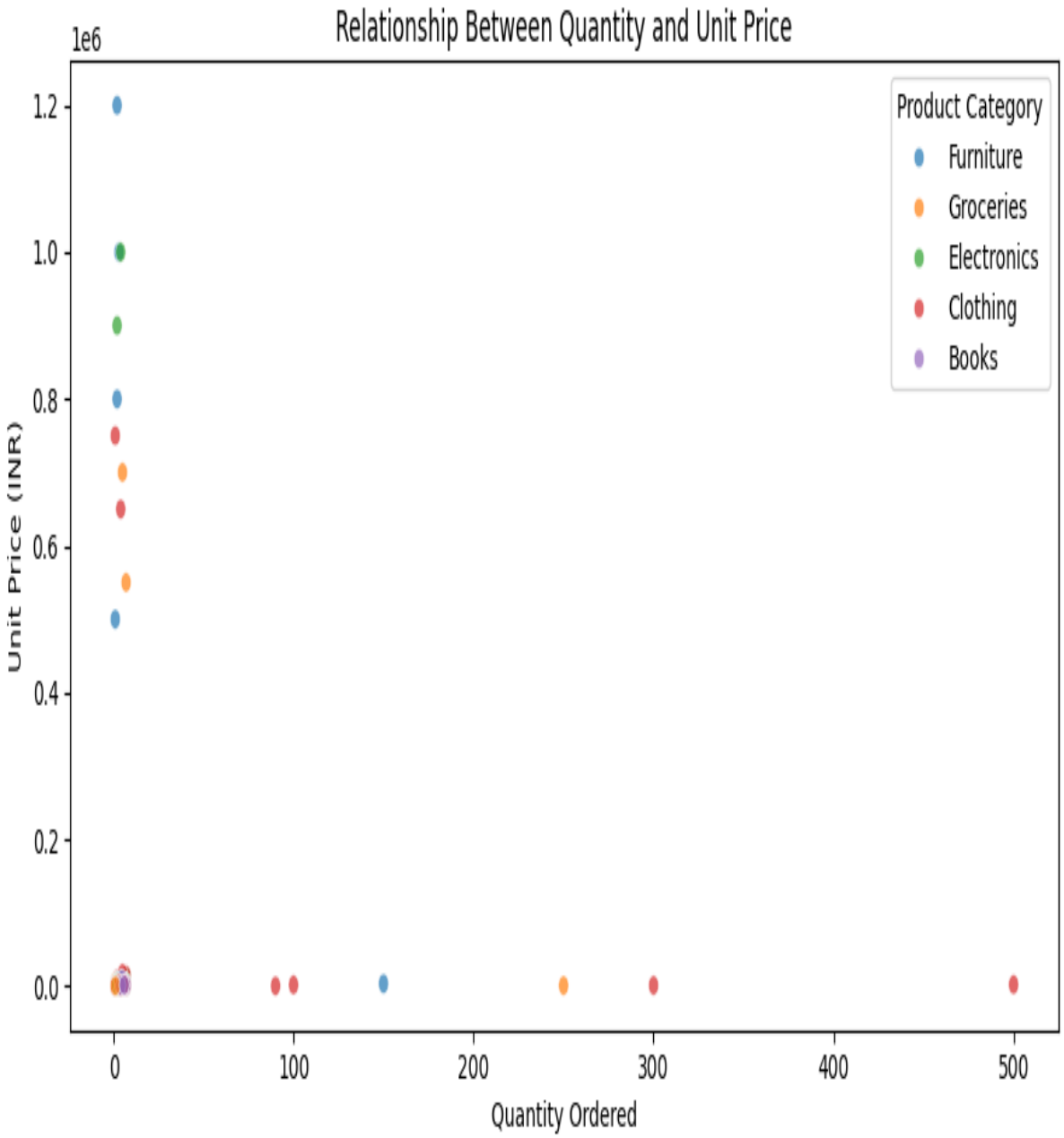
8. Monthly Trend Analysis



January 2025 was highest at about ■71.40 lakh; March was lowest at about ■3.81 lakh. July had the most orders (96) but about ■9.74 lakh gross value; December increased to about ■27.73 lakh. Extreme transactions may influence these totals. The graph does not prove seasonality or explain causes; 29 records without valid dates are excluded.

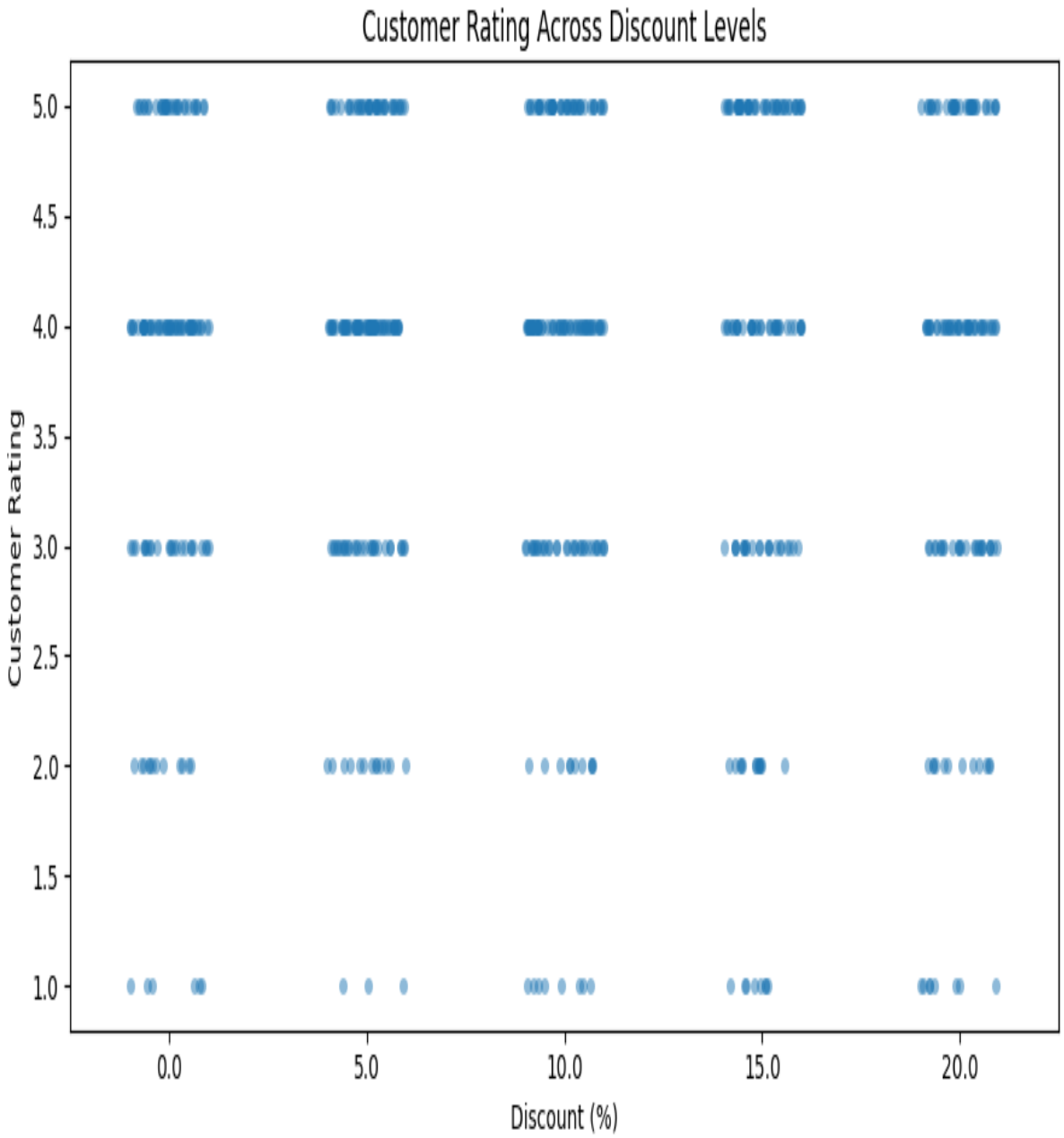
9. Scatter Plot Analysis

Quantity versus Unit Price



Most points cluster near the origin because extreme quantities/prices stretch the axes. No clear linear relationship is visible, but the scale limits interpretation.

Customer Rating Across Discount Levels



Ratings 1–5 appear at every discount level. Average ratings range from 3.68 at 20% discount to 3.83 at 5%; median rating is 4 in every group. This does not establish a causal effect.

Discount	Average Rating	Ratings Count	Median
0%	3.73	126	4.0
5%	3.83	145	4.0
10%	3.79	151	4.0
15%	3.80	125	4.0
20%	3.68	123	4.0

10. Sensitivity Analysis of Extreme Records

The screening rule flagged Quantity > 50 or Unit Price > ■10,000, identifying 20 of 1,000 orders.

Analysis	Number of Orders	Gross Order Value (INR)
All records	1,000	■32,101,023.94
Excluding flagged extremes	980	■5,467,789.88

Flagged records contribute ■26,633,234.06, approximately 83.0% of total gross order value while representing 2% of orders. This demonstrates sensitivity, not proof of error. The filtered total is a comparison, not corrected revenue.

11. Actionable Insights

1. Validate extreme transactions: Review the 20 flagged orders against source documents; correct only confirmed errors.

2. Compare category KPIs: Review order count, quantity, median price and validated net sales together.

3. Investigate monthly swings: Inspect January's high and March's low totals and re-check after validating extreme records.

4. Improve completeness: Prioritize missing ratings, dates and discounts; define the meaning of missing discount values.

5. Evaluate promotions holistically: Use margin, conversion and repeat purchases alongside ratings before changing discounts.

6. Improve chart readability: Use log scales or typical-range plots for highly skewed prices and quantities.

12. Limitations

• Extreme records have not been independently verified.
• Estimated value excludes missing discounts; gross value is before discounts.
• Missing ratings may bias rating analysis.
• Correlation does not establish causation.
• Monthly analysis excludes 29 records without valid dates.
• Results may not generalize beyond this dataset.

13. Conclusion

The EDA describes customer, order, price, discount, rating, category and monthly patterns. The selected numerical variables show weak linear correlations. The most important limitation is that 20 flagged records account for about 83% of gross order value. Verify these records before using totals for financial decisions. Better data completeness and separate full-range/typical-range reporting will improve reliability.